Weekly Update-2 [GROUP-6]

Tasks completed: MR-GPTQ + Simulated FP4 Microscaling Quantization on Edge Devices (MXFP4/NVFP4)

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Main findings of MR-GPTQ we tested:

- NVFP4 is generally more accurate than MXFP4 under naive round-to-nearest because NVFP4 uses smaller groups and more precise scales (E4M3).
- **Hadamard rotation normalizes ViT distributions just like LLMs**.
- Rotations/outlier mitigation behave differently: Hadamard rotations often help MXFP4 but can hurt NVFP4 under simple RTN.
- **MR-GPTQ**: GPTQ adapted to FP4 by combining (i) block-wise Hadamard 'micro-rotations', (ii) format-specific scale/grid optimization, and (iii) static (no-runtime-cost) activation reordering.
- **NVFP4 out performs MXFP4 but MR-GPTQ closes the gap for MXFP4.** MXFP4+MR-GPTQ achieved the highest accuracy of any method at 99.86%, surpassing even the FP32 baseline. Demonstrating that combining Hadamard rotation with GPTQ error compensation makes the hardware-friendlier MXFP4 format competitive.

Application Domain and What's Next

Since our target is an edge/mobile deployment (like a native app or MCUs) or a safety-critical detector:

1. Last week we started with INT8/INT4 baselines: confirm accuracy + latency + memory on BLIP-2 model.
2. Used GPTQ for weight-only INT4 first. Keeping activations higher precision (INT8/FP16) to reduce risk.
3. **This week** we extended microscaling FP4 (4-bit floating-point) quantization (originally developed for LLMs by Castro et al.[1]) to **vision transformers** for agricultural crop health monitoring, and deploy the resulting model to smartphones. [MCUs later on]
4. **Quantization experiments:** We implemented and compared 7 configurations: FP32 baseline, **INT4 RTN, NVFP4 RTN, MXFP4 RTN, MXFP4+Hadamard RTN, NVFP4+GPTQ, and MXFP4+MR-GPTQ (Hadamard rotation + GPTQ error**

compensation). All quantization methods maintained above 99.6% accuracy with near-perfect recovery rates.

5. Built a complete ONNX Runtime inference pipeline on CPU, replicating the exact preprocessing from training (resize to 256, center crop to 224, ImageNet normalization).
6. **Validation:** Confirmed 99.76% accuracy (2071/2076 correct) through ONNX Runtime, matching the PyTorch baseline. Only 5 misclassifications, all between visually similar tomato diseases.
7. Deployed a working Gradio web application accessible from a phone browser via public URL. The app accepts leaf photos, returns top-5 predictions with confidence scores, and provides disease-specific treatment recommendations. The model runs fully offline once loaded. No internet required for inference.

<u>Future Work for upcoming Weeks:</u>

we plan to add RAG systems to this although that raises concerns for MCUs. Explainability / Attention Visualization for Farmers, using VLMs. After it gives a diagnosis, show *where* the model is looking via attention rollout or GradCAM on the quantized ViT. We plan to look into: **does simulated FP4 quantization degrade attention map quality compared to FP16?**